SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA1305**

COURSE OUTCOME COVERED

CO1: Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**
Duration : 1 Hour

Analytical Problem

Code of Conduct:

*I **THIRUMALAPUDI PRANAVI (192411222)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

T. PRANAVI
Signature of the student:

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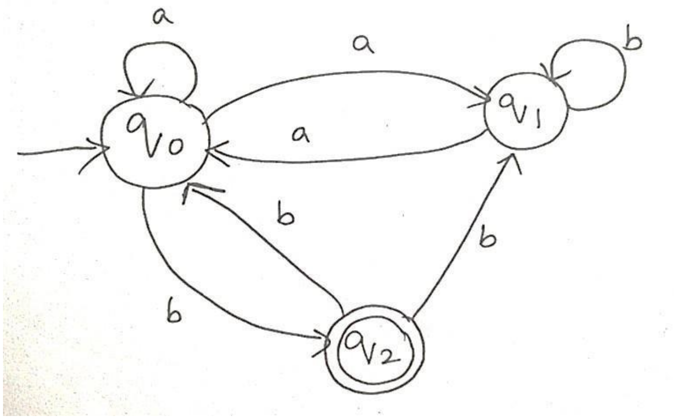
1. Consider the NFA given below which accepts some language L over the alphabet **Σ**={a,b}. Construct a DFA that represents the same language L

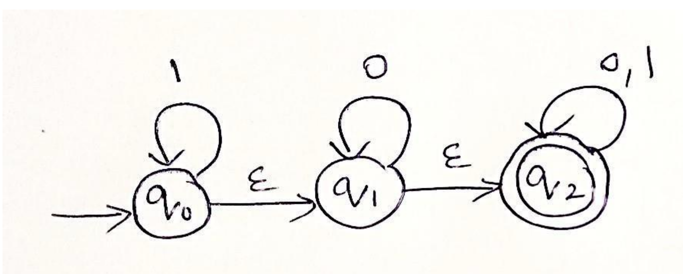


2. Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet **Σ**={0,1}. Construct a DFA that represents the same language.

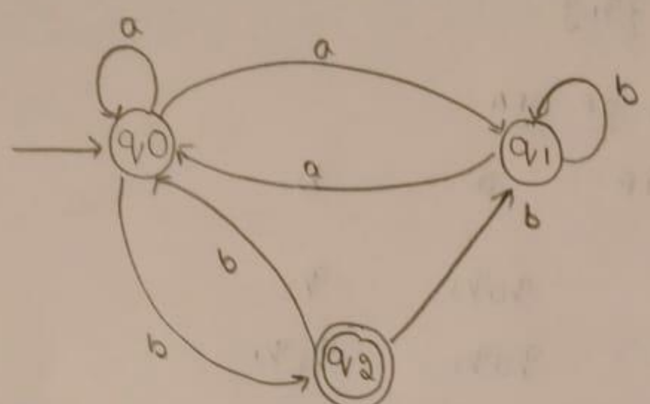
5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

1. Sol:



Given:

Alphabet $\Rightarrow$ $\Sigma = \{a, b\}$

Step 1: identify the NFA

States:

- q_0 (start state)
- q_1
- q_2 (Final state)

Final state:

$F = \{q_2\}$

Step 2: write the NFA Transition Table:

From the figure:

$\delta(q_0, a) = \{q_0 q_1\}$

$\delta(q_0, b) = \{q_2\}$

$\delta(q_2, a) = \{\emptyset\}$

$\delta(q_2, b) = \{q_0 q_1\}$

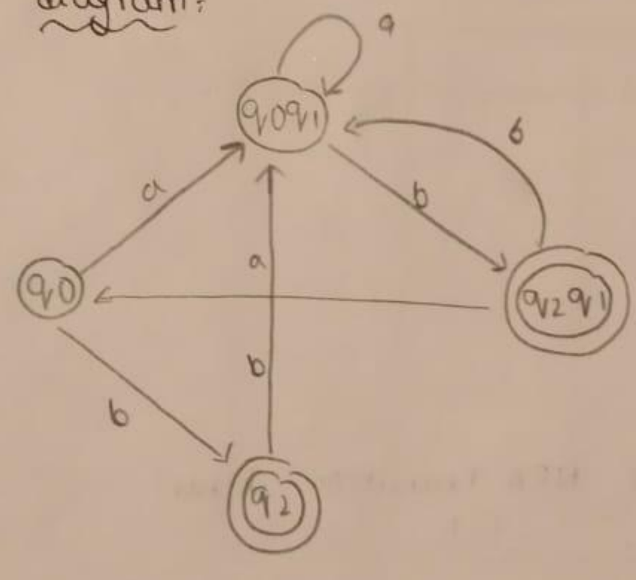
$\delta(q_1, a) = \{q_0\}$

$\delta(q_1, b) = \{q_1\}$

Transition Table of DFA

State / i/p	a	b
q_0	q_0q_1	q_2
q_0q_1	q_0q_1	q_2q_1
q_2	ϕ	q_0q_1
q_1q_2	q_0	q_0q_1

DFA diagram:



DFA = $\{Q, \Sigma, \delta, F, I\}$

$Q = \{q_0, q_0q_1, q_1q_2, q_2\}$

$\Sigma = \{a, b\}$

Initial state = $\{q_0\}$

final states = $\{q_2, q_1q_2\}$

$\delta(q_0, a) = \{q_0 q_1\}$

$\delta(q_0, b) = \{q_2\}$

$\delta(q_0q_1, a) = \{q_0q_1\}$

$\delta(q_0q_1, b) = \{q_1 q_2\}$

$\delta(q_1q_2, a) = \{q_0\}$

$\delta(q_1q_2, b) = \{q_0q_1\}$

$\delta(q_2, b) = \{q_0q_1\}$

2. Sol:

Step 1: NFA

States:

q_0 = start

q_1 = last symbol guessed as a

q_2 = last symbol guessed as b

q_3 = Final state

$M = \{Q, \Sigma, \delta, q_0, F\}$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{a, b\}$

Initial state = $\{q_0\}$

Final state = $\{q_3\}$

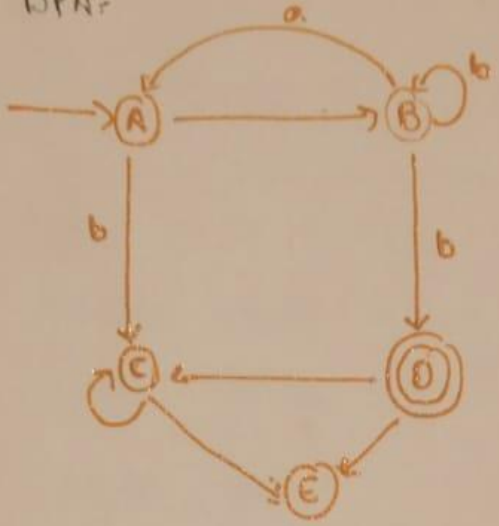
∴ The given string is abaab is accepted, bb is Rejected.

Step 2: DFA table & construction:

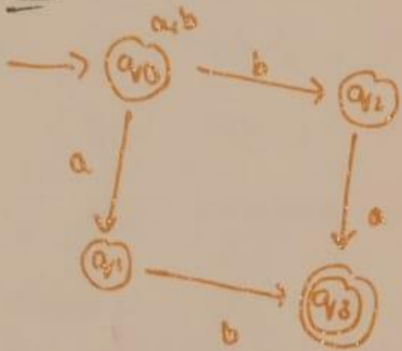
DFA State	a	b	let
A	B	C	$A = \{q_0\}$
B	B	D	$B = \{q_0, q_2\}$
C	E	C	$C = \{q_0, q_2\}$
*D	E	C	$D = \{q_0, q_2, q_3\}$
*E	B	D	$E = \{q_0, q_1, q_3\}$

Final states : D, E

DFA:-



NFA :



3. Sol:

Formula defination;

The NFA is

$M = (\emptyset, \epsilon, G, q_0, F)$

So,

Now, current states (q_0, q_2)

Read 1:

From $q_0 \rightarrow (q_0, q_2)$

From $q_2 \rightarrow (q_3)$

∴ current state = (q_0, q_1, q_3)

Finally,

Start state = (q_0)

Final state = (q_3)

Formal definition

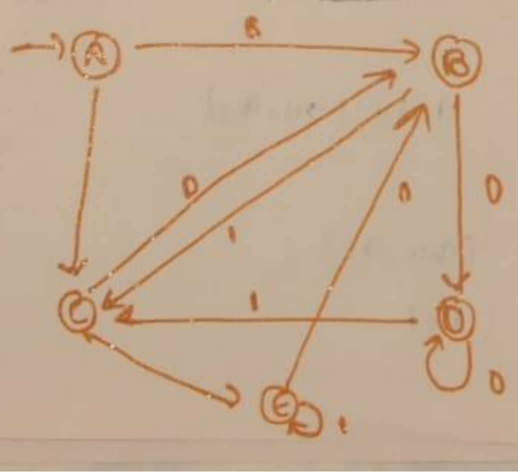
Q: $q_0 q_1, q_2, q_3$

$\Sigma = (0, 1)$

δ as given in transition table.

DFA State:

DFA	0	1
A	B	C
B	D	C
C	B	E
*D	D	E
*E	B	E



4. Sol:

A The alphabet is $\Sigma = \{0, 1\}$

The DFA should accept all binary strings containing 001 as a substring

Accepted strings: 001, 1001, 0001, 0010, 11001, 10010

Rejected strings: ϵ, 0, 1, 00, 011, 1111, 01010

Step 2: DFA

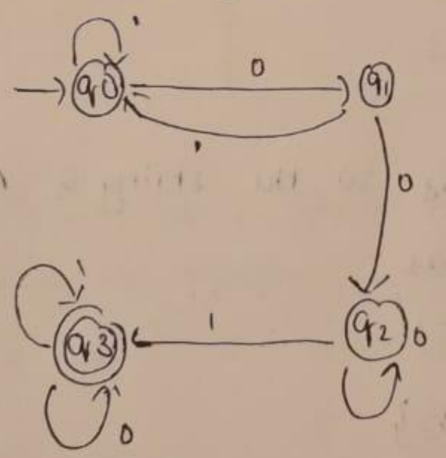
q_0 : start state

q_1 : matched 0

q_2 : matched 00

q_3 : matched 001 (Final state)

Step 3: State diagram



Transition table:

present state	input 0	input 1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$\rightarrow q_3$	q_3	q_3

Check with an example string = 10010

Input	current state
initial	q_0
1	q_0
0	q_1
0	q_2
1	q_3
0	q_3

The machine ends in q_3 so the string is Accepted

The DFA is defined as

$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

q_0 = start state

$F = \{q_3\}$

Transition function:

δ	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
q_3	q_3	q_4

∴ The DFA has 4 states

∴ it accepts all binary strings containing the substring 001.

Start state: q_0

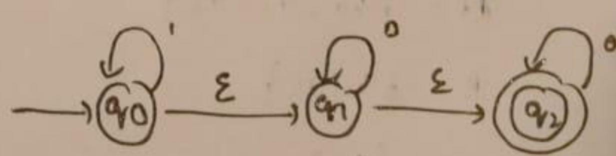
Final state: q_3

once the substring 001 is detected, the DFA remains in the accepting state for all meaning inputs

5. <u>Sol:</u>

The ε-closure at a state is the set of all states that can be reached from the state by taking zero or more ε-transitions

$q_0 \xrightarrow{\varepsilon} q_1$
$q_1 \xrightarrow{\varepsilon} q_2$



The ε-closure (q_0) = $\{q_0, q_1, q_2\}$

because q_0 can reach q_1 and q_2 using only ε-transitions

suppose the given NFA has the following ε-transitions.

$q_0 \rightarrow \varepsilon q_1$

$q_1 \rightarrow \varepsilon q_2$

ε-closure (q_0)

start from q_0

reach q_1 through ε.

From q_1 reach q_2

ε-closure (q_0) = $\{q_0, q_1, q_2\}$

ε-closure (q_1) = $\{q_1, q_2\}$

ε-closure (q_2) = $\{q_2\}$

Step 3: construct the DFA

The DFA states are the ε-closures or subsets) of the NFA states

$A = \{q_0, q_1, q_2\}$

$B = \{q_1, q_2\}$

$C = \{q_2\}$

$D = \phi$ Dead state

Step 4: using the subset construction method

From A = $\{q_0, q_1, q_2\}$

on input 0

move To B

on input 1

Remain in A

From B = $\{q_1, q_2\}$

on 0

Remain in B

on 1

move to C

From C = $\{q_2\}$

on 0

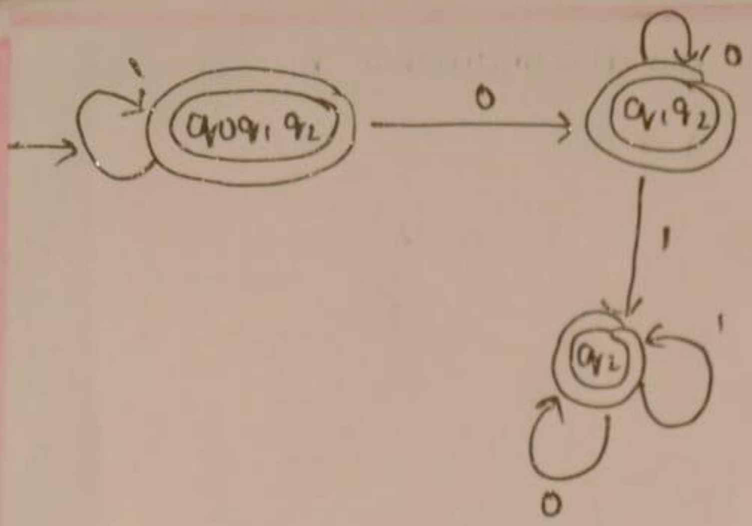
Remain in C

on 1

Remain in C

Dead state : Both inputs return to the dead state

DFA state	0	1
A = $\{q_0, q_1, q_2\}$	B	A
B = $\{q_1, q_2\}$	B	C
C = $\{q_2\}$	C	C
D = $\emptyset$	D	D



initial state = {A}

final state = {A B C}

$\therefore$ A = $\{q_0, q_1 q_2\}$

B = $\{q_1 q_2\}$

C = $\{q_2\}$

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